

REQUEST FOR INFORMATION (RFI) 70RWMD26RFI00000001

Title: Environmental Microbial Microflora Differentiation Between Cities

NOTE: THIS IS A REQUEST FOR INFORMATION (RFI) ONLY. This RFI is issued solely for information and market research planning purposes. This RFI shall not be construed as an obligation on the part of the Government. This is NOT a Request for Quotations or Proposals. No solicitation document exists, and the Government may or may not issue a formal solicitation as a result of the responses received to this RFI.

The Government will not pay for any response or demonstration expenses. All costs incurred responding to this RFI will be solely at the interested party's expense. Failure to respond to this RFI will not preclude participation in any future solicitation. Any information received will become the property of the Government and will not be returned to the submitter. Interested parties are responsible for adequately marking proprietary or sensitive information.

The Government's explicit intent, in this request for information, is to not receive from respondents any proprietary data, trade secrets, business sensitive information, or information considered CONFIDENTIAL under 18 U.S.C. §1905. The Government's constraint does not in any way relieve contractors from their responsibility to properly mark proprietary data when it is provided, nor does it alleviate any requirement for the Government to protect marked data. The Government is not obligated to protect unmarked data. The information provided in this RFI is subject to change and is not binding on the Government. All submissions become the property of the Federal Government and will not be returned.

DESCRIPTION

The Department of Homeland Security (DHS) Countering Weapons of Mass Destruction Office's (CWMD) mission is to enable operational partners to prevent weapons of mass destruction use against the Homeland and promote readiness for chemical, biological, radiological, and nuclear threats. We have improved biopreparedness and situational awareness by sharing public health and intelligence information with state and local partners through local fusion centers. These measures provide expert analysis of national security vulnerabilities and how to remedy them, and funding to prepare the state and local law enforcement who will respond first to any biological threats.

DHS is committed to using cutting edge technologies and scientific talent in our quest to make America safer. Therefore, CWMD has a mission to obtain technological resources and tools that are reliable and available for use, as well as leveraging existing resources to help protect the homeland.

PURPOSE OF THIS REQUEST FOR INFORMATION (RFI)

The purpose of this RFI is to collate information on potential partners, technologies, and/or methods that could be used to enhance the Federal Government's understanding of the presence, identity, and abundance of microbes that may create background noise, distortion, delay, or general interference in future agnostic (i.e., non-species specific) detection/identification via genetic sequencing / metagenomic screening efforts for pathogens of national security concern. CWMD may seek to employ organizations and/or methods identified through this RFI in future research and development (R&D) efforts to address the improvement and efficiency of biosurveillance across the U.S., and to better-understand what

microbial diversity may exist locally or nationally that could confuse, slow, or disrupt our efforts and instrumentation.

Recognizing the critical need for real-time detection and identification of aerosolized or airborne biological agents, CWMD is seeking responses through this Request for Information (RFI) on capabilities available through private industry and academia that could be leveraged to improve the biothreat detection and identification accuracy of disease diagnostics conducted across the U.S., specifically by metagenomically characterizing background microbial microflora and fauna profiles in locations served by the BioWatch program, major cities, or other sites of current or potential domestic biodefense activities.

The DHS CWMD Office wants to enhance their understanding of what methods and materiel solutions exist and can meet these needs regardless of their current technology readiness: from basic through applied research efforts. In addition, the CWMD Office seeks to gain an understanding with a Rough Order of Magnitude (ROM) cost estimate and a general timeline duration.

INFORMATION REQUEST

Interested parties should submit responses addressing the following:

1. More specifically, respondents should communicate how they could assist CWMD in performance of the following: Identification and analysis of knowledge gaps in metagenomic sequencing data for domestic microbial microflora and fauna.
2. Benefits to future biothreat detection and identification that would be provided by a metagenomic sequencing approach to microflora and fauna differentiation between US cities (primarily BioWatch locations);
3. Plan and execute development of ‘metagenomic ‘baselines’ in select US cities that address geographical, seasonal, diurnal/nocturnal, local weather conditions including temperature, humidity, solar radiation, and precipitation;
4. Analysis of collected environmental background data to determine trends and how different microflora and fauna may impact the effectiveness current and future PCR- and sequencing-based biological threat detection activities in cities across the US.

Sources are further encouraged to consider existing methods of environmental metagenomic sequencing, modalities of airborne or aerosolized biothreat detection, and suggestions on how future CWMD efforts could improve upon the status quo of biothreat surveillance within the U.S. in their RFI response.

RESPONSES TO RFI

Interested parties should submit electronic responses to the point(s) of contact identified below:

Ariana Thompson, Contract Specialist, Ariana.Thompson@hq.dhs.gov.

RFI responses should be in .docx format. Figure legends, labels, and table of contents may be no smaller than font size 8 as presented in Microsoft Word. Responses shall be limited to 10 pages inclusive of cover page containing respondent information. Proprietary information, if any, should be minimized and **MUST BE CLEARLY MARKED**. The subject line of the email should contain the RFI number (70RWMD26RFI00000001) and “Environmental Microbial Microflora Differentiation Between Cities.”

At a minimum, responses should include:

1. Company information: (name, phone number, UEI, CAGE Code, POC information, website, etc.).
2. GSA contract(s): contract numbers and contractor's socioeconomic status.
3. Description of your company's capabilities as they relate to the services described in this notice.
4. A description of your company's previous experience providing the services described in this notice, including independent testing that substantiates your capabilities.
5. A description of any teaming/subcontracting your company anticipates utilizing in support of this requirement.
6. A statement indicating the ability of your company to meet the proposed period of performance (POP) OR a detailed explanation of what a feasible POP to complete this effort would look like.
7. A proposed method, timeline and cost estimate to complete the above-described research and development effort.
8. Any other relevant information that is not listed above which the Government should consider in finalizing its market research and acquisition strategy.